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Aim : Aim: Wireless Sensor Network Deployment and Coverage Analysis
Simulate random and grid deployment of sensor nodes and evaluate network coverage and connectivity.

```
# =====  
# WSN Grid Deployment and Coverage Analysis  
# NS-2.35 Compatible  
# =====  
  
# -----  
# Create Simulator  
# -----  
  
set ns [new Simulator]  
  
# -----  
# Trace Files  
# -----  
  
set tracefile [open wsn.tr w]  
$ns trace-all $tracefile  
  
set namfile [open wsn.nam w]  
$ns namtrace-all-wireless $namfile 500 500  
  
# -----  
# Topography  
# -----  
  
set topo [new Topography]  
$topo load_flatgrid 500 500  
  
# -----  
# Simulation Parameters  
# -----  
  
set val(chan) Channel/WirelessChannel  
set val(prop) Propagation/TwoRayGround  
set val(netif) Phy/WirelessPhy  
set val(mac) Mac/802_11  
set val(ifq) Queue/DropTail/PriQueue  
set val(ll) LL  
set val(ant) Antenna/OmniAntenna  
set val(ifqlen) 50
```

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```
set val(nn)      9
set val(rp)      AODV
```

```
# -----
# Create GOD
# -----
```

```
create-god $val(nn)
```

```
# -----
# Wireless Channel
# -----
```

```
set chan_1_ [new $val(chan)]
```

```
# -----
# Configure Nodes
# -----
```

```
$ns node-config \
  -adhocRouting $val(rp) \
  -llType $val(ll) \
  -macType $val(mac) \
  -ifqType $val(ifq) \
  -ifqLen $val(ifqlen) \
  -antType $val(ant) \
  -propType $val(prop) \
  -phyType $val(netif) \
  -channel $chan_1_ \
  -topoInstance $topo \
  -agentTrace ON \
  -routerTrace ON \
  -macTrace ON
```

```
# -----
# Create 9 Sensor Nodes
# -----
```

```
for {set i 0} {$i < $val(nn)} {incr i} {
  set node($i)[ns node]
}
```

```
# =====
# GRID DEPLOYMENT (3 x 3)
```

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=====

Row 1

\$node(0) set X_ 100

\$node(0) set Y_ 100

\$node(0) set Z_ 0

\$node(1) set X_ 200

\$node(1) set Y_ 100

\$node(1) set Z_ 0

\$node(2) set X_ 300

\$node(2) set Y_ 100

\$node(2) set Z_ 0

Row 2

\$node(3) set X_ 100

\$node(3) set Y_ 200

\$node(3) set Z_ 0

\$node(4) set X_ 200

\$node(4) set Y_ 200

\$node(4) set Z_ 0

\$node(5) set X_ 300

\$node(5) set Y_ 200

\$node(5) set Z_ 0

Row 3

\$node(6) set X_ 100

\$node(6) set Y_ 300

\$node(6) set Z_ 0

\$node(7) set X_ 200

\$node(7) set Y_ 300

\$node(7) set Z_ 0

\$node(8) set X_ 300

\$node(8) set Y_ 300

\$node(8) set Z_ 0

Set Initial Node Positions

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```
for {set i 0} {$i < $val(nn)} {incr i} {
    $ns initial_node_pos node(i) 20
}

# =====
# TRAFFIC
# Source Node 0 -> Destination Node 8
# =====

# UDP Agent
set udp0 [new Agent/UDP]
$ns attach-agent $node(0) $udp0

# Null Agent
set null0 [new Agent/Null]
$ns attach-agent $node(8) $null0

# Connect source and destination
$ns connect $udp0 $null0

# CBR Traffic
set cbr0 [new Application/Traffic/CBR]

$cbr0 set packetSize_ 512
$cbr0 set interval_ 0.1

$cbr0 attach-agent $udp0

# -----
# Start and Stop Traffic
# -----

nsat1.0 "cbr0 start"
nsat8.0 "cbr0 stop"

# =====
# FINISH PROCEDURE
# =====

proc finish {} {
    global ns tracefile namfile

    $ns flush-trace
```

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```
close $tracefile
close $namfile

puts "Simulation Completed"
puts "Trace file: wsn.tr"
puts "NAM file: wsn.nam"

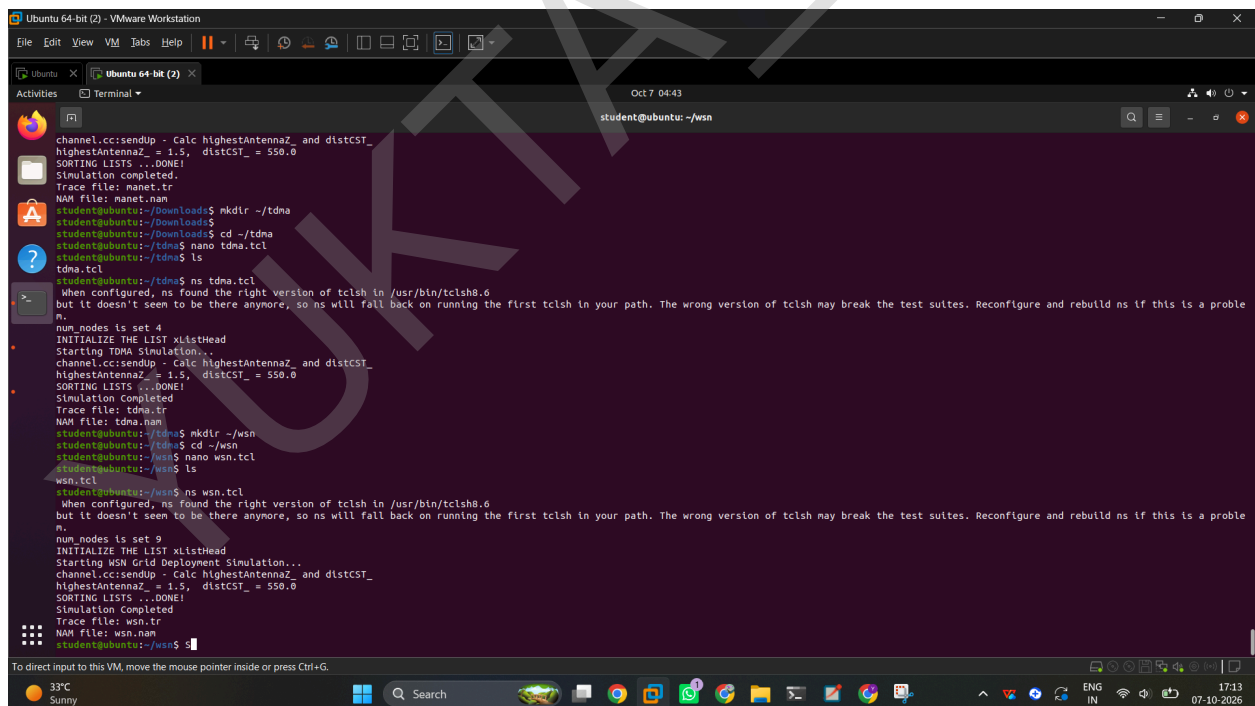
exec nam wsn.nam &
exit 0
}

# -----
# End Simulation
# -----

$ns at 10.0 "finish"

puts "Starting WSN Grid Deployment Simulation..."

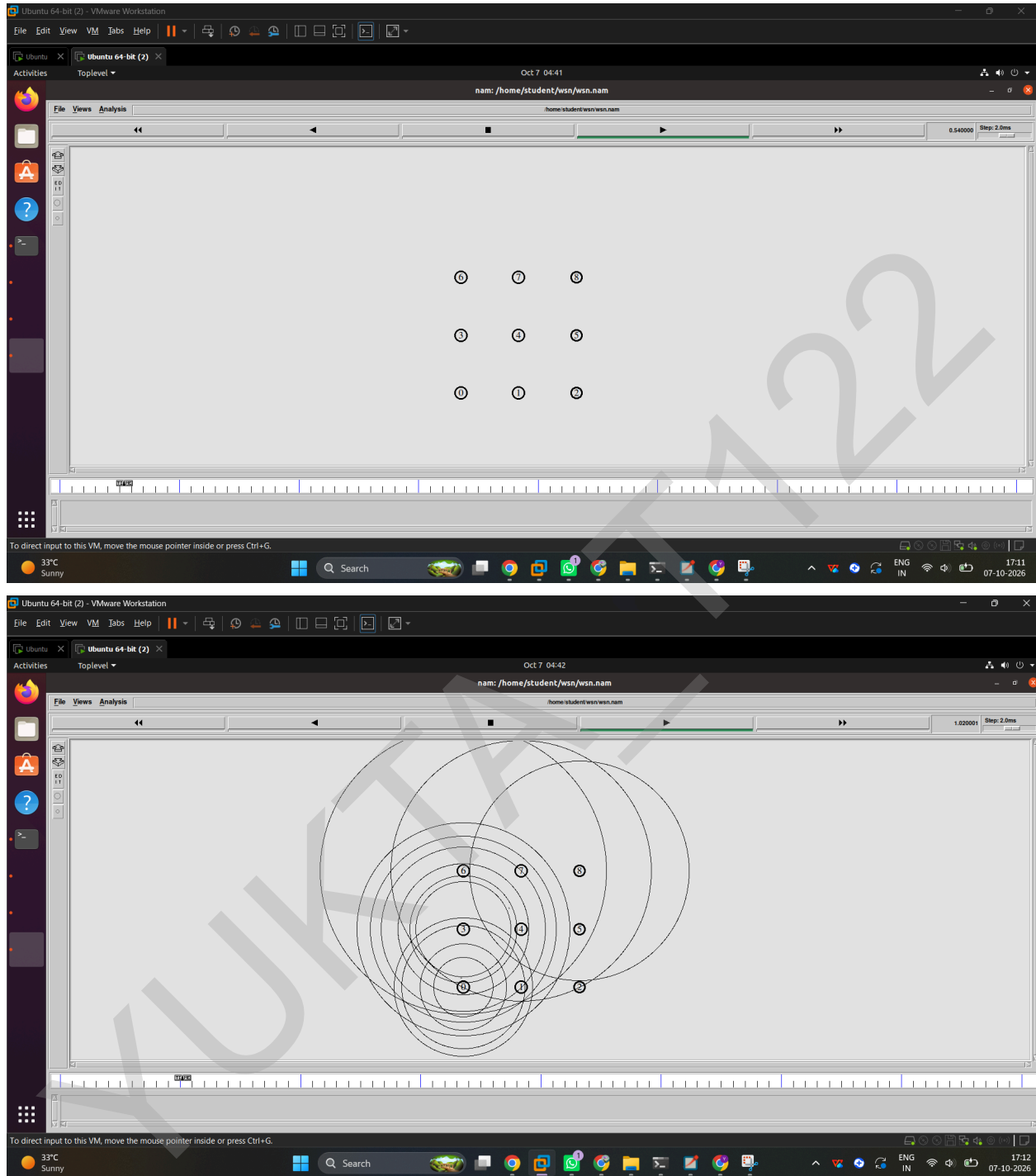
$ns run
```



```
Ubuntu 64-bit (2) - VMware Workstation
File Edit View VM Tabs Help
Oct 7 04:43
student@ubuntu: ~/wsn

channel.cc:sendup - Calc highestAntennaZ_ and distcst_
highestAntennaZ_ = 1.5, distcst_ = 550.0
SORTING LISTS ...DONE!
Simulation Completed.
Trace file: wsn.tr
NAM file: wsn.nam
student@ubuntu:~/Downloads$ mkdir ~/tdna
student@ubuntu:~/Downloads$ cd ~/tdna
student@ubuntu:~/tdna$ nano tdna.tcl
student@ubuntu:~/tdna$ ls
tdna.tcl
student@ubuntu:~/tdna$ ns tdna.tcl
When configured, ns found the right version of tcclsh in /usr/bin/tcclsh8.6
but it doesn't seem to be there anymore, so ns will fall back on running the first tcclsh in your path. The wrong version of tcclsh may break the test suites. Reconfigure and rebuild ns if this is a problem.
num_nodes is set 4
INITIALIZE THE LIST xlistHead
Starting TDMA Simulation...
channel.cc:sendup - Calc highestAntennaZ_ and distcst_
highestAntennaZ_ = 1.5, distcst_ = 550.0
SORTING LISTS ...DONE!
Simulation Completed
Trace file: tdna.tr
NAM file: tdna.nam
student@ubuntu:~/tdna$ mkdir ~/wsn
student@ubuntu:~/tdna$ cd ~/wsn
student@ubuntu:~/wsn$ nano wsn.tcl
student@ubuntu:~/wsn$ ls
wsn.tcl
student@ubuntu:~/wsn$ ns wsn.tcl
When configured, ns found the right version of tcclsh in /usr/bin/tcclsh8.6
but it doesn't seem to be there anymore, so ns will fall back on running the first tcclsh in your path. The wrong version of tcclsh may break the test suites. Reconfigure and rebuild ns if this is a problem.
num_nodes is set 9
INITIALIZE THE LIST xlistHead
Starting WSN Grid Deployment Simulation...
channel.cc:sendup - Calc highestAntennaZ_ and distcst_
highestAntennaZ_ = 1.5, distcst_ = 550.0
SORTING LISTS ...DONE!
Simulation Completed
Trace file: wsn.tr
NAM file: wsn.nam
student@ubuntu:~/wsn$
```

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